

XGBoost Fraud Model - Test-Set Performance by Threshold

Final model: XGBoost (Batch 3, regularization-refined) | ROC-AUC and PR-AUC are threshold-independent

Operating point	Threshold	Precision	Recall	F1	ROC-AUC	PR-AUC
0.5 (precision-leaning)	0.630	0.951	0.578	0.719	0.9469	0.8335
F1 (balanced)	0.338	0.832	0.685	0.752	0.9469	0.8335
F1.3	0.276	0.785	0.714	0.748	0.9469	0.8335
F1.5 (~crossover)	0.231	0.740	0.746	0.743	0.9469	0.8335
F1.75 <-- CHOSEN	0.195	0.691	0.769	0.728	0.9469	0.8335
F2 (recall-leaning)	0.171	0.648	0.785	0.710	0.9469	0.8335
Default 0.5 (untuned)	0.500	0.907	0.633	0.746	0.9469	0.8335

Sweet spot: F1.75-optimal threshold (0.195) is the chosen production operating point - recall (0.769) is clearly ahead of precision (0.691), per the business priority of catching fraud, while precision stays well above the more extreme F2 point (0.649). Thresholds are chosen from out-of-fold training predictions, confirmed on validation, then evaluated once on this held-out test set.